Article

AERIS: Environment-Aware Hierarchical Routing for Reliable Wireless Sensor Networks under Realistic Channels

Xiaobo Zhang [1], Kangrui Li [1,*] and Junyi Lin [1]

[1] Faculty of Automation, Guangdong University of Technology, Guangzhou 510006, China
* Correspondence: 1403073295@mails.gdut.edu.cn

Version February 14, 2026 submitted to Sensors

Abstract: This paper evaluates AERIS, a lightweight hierarchical routing protocol for wireless sensor networks under heterogeneous channel conditions. The primary metric is `pdr_expected` (base-station delivered packets divided by expected source packets). In the publication-tier 100-node matrix (four environments, 30 seeds per environment), AERIS achieves the highest packet delivery ratio within the tested baseline set (LEACH, PEGASIS, HEED, and TEEN). In the large-scale matrix (100–1000 nodes; indoor factory n=650, other environments n=550), AERIS ranks first in three environments (indoor factory, outdoor urban, and outdoor suburban), while PEGASIS remains higher in indoor office. Ablation indicates environment-dependent module contribution: Gateway is positive in three environments and near-neutral in indoor office; CAS does not provide a consistent positive marginal effect. Hop-based latency analysis shows AERIS near two hops, while PEGASIS remains above 31 hops due to chain traversal. NS-3 evidence is used strictly as trend-level cross-platform validation (AERIS vs LEACH), not as a numerical-equivalence proof. The final claim is bounded to the tested protocols, environments, metrics, and sample-size regimes.

Keywords: wireless sensor networks; reliable routing; hierarchical protocol; packet delivery ratio; scalability; reproducible evaluation

---

1. Introduction

Wireless sensor network (WSN) routing is classically optimized for energy reduction, but practical deployments also require stable delivery under heterogeneous channels [1–3]. This creates a multi-objective tension among reliability, hop-based latency, and energy use. Existing protocols represent distinct operating points: LEACH emphasizes clustered forwarding [4], PEGASIS emphasizes chain-based aggregation [5], HEED emphasizes residual-energy-aware head selection [6], and TEEN emphasizes event-triggered reporting [7]. Prior channel studies also show that low-power links can be asymmetric and environment-sensitive under realistic propagation conditions [8–10].

AERIS is designed as a lightweight hierarchical protocol that prioritizes reliability under realistic channel variability. The design objective is not universal dominance, but robust performance across multiple channel environments while maintaining deployability on resource-constrained nodes.

Contributions and Scope Boundaries

This manuscript makes three scoped contributions. First, it provides publication-tier evidence for multi-environment reliability comparison at 100 nodes and environment-stratified scalability analysis up to 1000 nodes. Second, it quantifies module-level behavior with ablation under the same channel taxonomy. Third, it adds cross-platform trend validation in NS-3 for AERIS versus LEACH. The manuscript explicitly does not claim universal protocol superiority or cross-platform numerical equivalence. Every inference is constrained to the reported metric, environment set, and sample-size regime.

2. Related Work

Classical WSN protocols (LEACH, PEGASIS, HEED, TEEN) provide foundational routing mechanisms but show different failure modes under harsh channels and larger scales [3,5,6,11]. Chain-based aggregation can reduce per-hop transmission effort but increases end-to-end hop count. Cluster-based approaches can reduce hop count but may become sensitive to uplink quality and head placement.

Recent adaptive or learning-based methods improve specific settings but often require heavier runtime or training overhead [12–14]. Context-aware routing studies further motivate explicit environmental adaptation in protocol design [15,16]. In this manuscript, the focus is a rule-based protocol with explicit reproducibility constraints and auditable experiment metadata.

3. System Model and Protocol

3.1. Metric Definition

The primary metric is

$$\mathrm{PDR}_{\mathrm{expected}} = \frac{\mathrm{bs\_delivered}}{\mathrm{source\_packets\_expected}}. \tag{1}$$

All publication conclusions in this draft are based on this metric.

3.2. AERIS Architecture

AERIS combines three components:

1. Context-Adaptive Switching (CAS) for mode control,
2. Skeleton backbone for hierarchical forwarding,
3. Gateway coordination for CH-to-BS reinforcement.

Gateway candidate scoring follows a weighted rule

$$\mathrm{Score}_i = \alpha E_i + \beta C_i + \gamma L_i \tag{2}$$

where E_i is residual energy, C_i is centrality, and L_i is link quality.

4. Experimental Setup

4.1. Publication-Tier Settings

- Simulator: Python implementation with logged provenance.
- Channel environments: indoor_office, indoor_factory, outdoor_urban, outdoor_suburban.
- Default node count: 100.
- Seeds for multi-environment and ablation: 30 independent seeds.
- Seeds for scalability: environment-specific sample sizes with n=550 (indoor factory uses n=650).
- Baselines: LEACH, PEGASIS, HEED, TEEN.

4.2. Evidence Scope

The analysis is grounded in publication-tier multi-environment comparison, ablation, scalability, hop-based latency, energy-lifetime statistics, and NS-3 trend validation. Claims are constrained to this evidence scope. Cross-platform numerical equivalence is intentionally not claimed, and cross-regime numerical pooling is avoided.

4.3. Statistical Methods

For pairwise protocol comparisons, we use Welch's two-sample t-test because variance equality is not assumed across protocols and environments. Family-wise control is handled by Holm correction over each comparison family. Effect size is reported with Hedges' g and interpreted as: small (0.2), medium (0.5), and large (0.8) in absolute value. Confidence intervals are reported as two-sided 95% intervals. All significance statements in this manuscript are based on Holm-corrected p values.

4.4. Reproducibility Controls

All publication-tier scripts in this study use deterministic seed lists. They also enforce `force_ctp_reliable=False`. This guarantees that PDR reflects simulated delivery behavior rather than any reliability override. We report `pdr_expected` as the primary metric and treat hop count as a latency proxy, not a wall-clock latency measurement.

5. Results

5.1. Multi-Environment Comparison at 100 Nodes (n=30)

Table 1. PDR by environment at 100 nodes (mean ± std; std follows the result-file convention with ddof=0).

Environment	AERIS	LEACH	PEGASIS	HEED	TEEN
indoor_office	0.9739±0.0047	0.5543±0.0401	0.9078±0.0166	0.9371±0.0076	0.8222±0.0044
indoor_factory	0.6031±0.0258	0.1614±0.0209	0.1928±0.0255	0.2326±0.0263	0.3113±0.0245
outdoor_urban	0.3745±0.0354	0.0552±0.0127	0.0542±0.0117	0.0635±0.0121	0.1201±0.0183
outdoor_suburban	0.7451±0.0193	0.2703±0.0272	0.3382±0.0329	0.4221±0.0313	0.4752±0.0236

At 100 nodes, AERIS ranks first in all four tested environments within the evaluated baseline set (LEACH, PEGASIS, HEED, TEEN). This claim is restricted to the 100-node matrix and should not be generalized to the 1000-node regime.

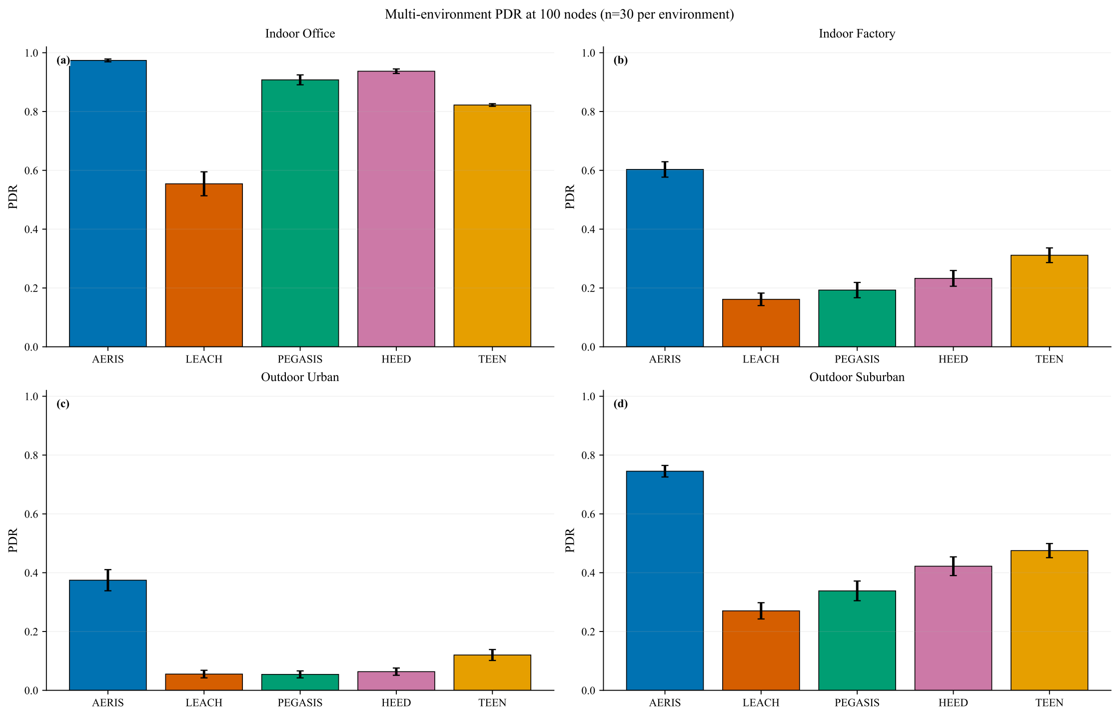

Figure 1. PDR comparison panel at 100 nodes (4 environments, n=30).

5.2. *Ablation: Environment-Dependent Effects (n=30)*

Table 2. Full vs no_gateway PDR (mean ± std; std reported as population standard deviation, ddof=0).

Environment	Full	no_gateway	Delta (no_gateway - full)
indoor_office	0.9739±0.0047	0.9741±0.0036	+0.0002
indoor_factory	0.6031±0.0258	0.5806±0.0215	-0.0225
outdoor_urban	0.3745±0.0354	0.3534±0.0301	-0.0212
outdoor_suburban	0.7451±0.0193	0.7306±0.0264	-0.0146

Gateway is positive in three environments and near-neutral in indoor_office. CAS is not consistently positive across environments. In indoor_office, the Full-vs-no_gateway gap is +0.0002 (absolute PDR), so this cell is treated as practically neutral.

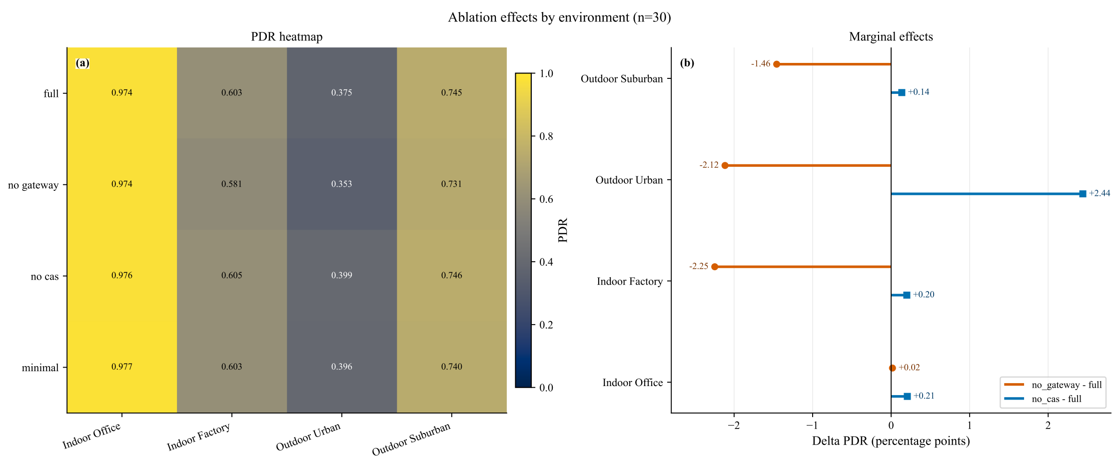

Figure 2. Ablation panel: heatmap and marginal effects (n=30).

5.3. Scalability (100–1000 Nodes; n=550/650)

Table 3. PDR ranking at 1000 nodes (mixed sample sizes: n=650 in indoor_factory, n=550 in other environments).

Environment	AERIS	LEACH	PEGASIS	HEED	TEEN	AERIS rank
indoor_office	0.9899	0.9902	0.9991	0.9911	0.9922	5th
indoor_factory	0.9725	0.1888	0.1619	0.1631	0.2196	1st
outdoor_urban	0.8849	0.0623	0.0487	0.0341	0.0664	1st
outdoor_suburban	0.9900	0.5952	0.7866	0.5544	0.6899	1st

AERIS is first in 3/4 environments in the large-scale matrix, while PEGASIS is higher in indoor_office. This statement is matrix-specific and must not be interpreted as a pooled cross-environment ranking. Because sample sizes differ by environment (n=550 or n=650), all large-scale claims are reported per environment and are not pooled into a single global estimate.

5.4. Statistical Robustness Snapshot (AERIS vs LEACH)

Table 4. Representative Holm-corrected significance snapshot from scalability matrix.

Environment	Nodes	ΔPDR	Hedges' g	Holm p	Significant
indoor_office	100	+0.004548	+2.3398	1.32×10^{-207}	yes
indoor_office	500	-0.000130	-0.1009	9.46×10^{-2}	no
indoor_office	1000	-0.000286	-0.2673	1.05×10^{-5}	yes
indoor_factory	1000	+0.783773	+151.3826	$< 10^{-300}$	yes
outdoor_urban	1000	+0.822585	+181.4951	$< 10^{-300}$	yes
outdoor_suburban	1000	+0.394856	+65.3769	$< 10^{-300}$	yes

This snapshot highlights the scoped conclusion used throughout this manuscript: strong and consistent gains in harsh environments, but environment-dependent behavior in benign indoor_office conditions. For very large sample sizes, tiny mean differences can still produce small p values; therefore, practical interpretation in this study follows both effect size and absolute PDR gap, not significance labels alone (for example, indoor_office at 1000 nodes has a small negative delta versus LEACH). The very large absolute values of Hedges' g in harsh environments reflect both large mean gaps and low within-group variance under large sample sizes; they should be interpreted as magnitude indicators within this experiment design, not as cross-study constants.

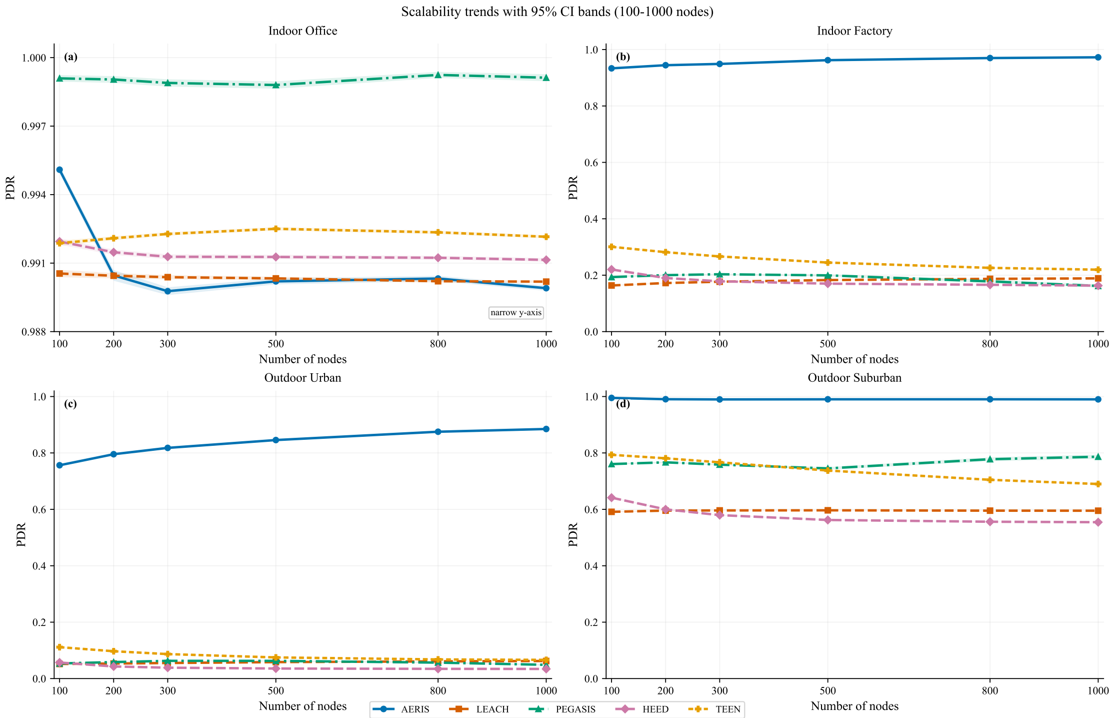

Figure 3. Scalability trends over node counts (indoor_factory n=650, other environments n=550) with 95% CI bands. The indoor_office panel uses a narrowed y-axis window to expose small but non-zero ordering gaps near PDR=1.0.

103 ### 5.5. Hop-Based Latency and Trade-offs

104 At 100 nodes (n=30), AERIS stays near two hops (1.97–1.99), while PEGASIS remains above 31
105 hops due to chain traversal. LEACH and TEEN show lower average hops because of protocol paths
106 that allow direct-to-BS transmissions.

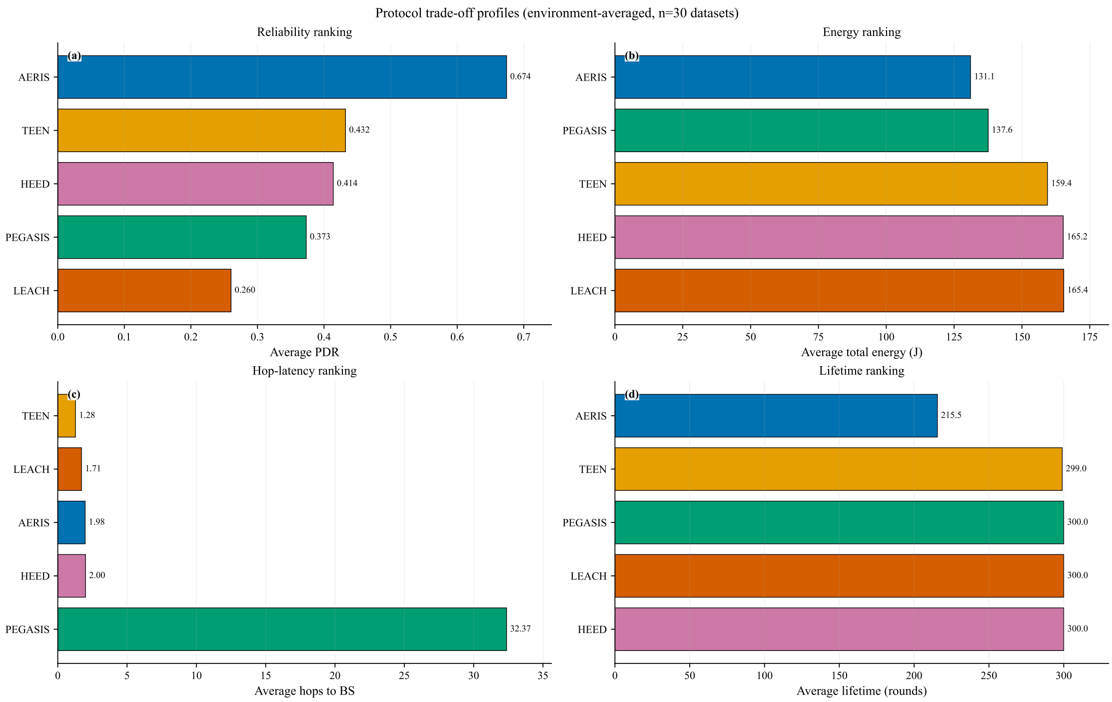

Figure 4. Trade-off panel from publication-tier data: reliability, energy, hop-based latency, and lifetime.

5.6. NS-3 Trend Validation (AERIS vs LEACH, n=30)

NS-3 validation is used as a cross-platform trend check rather than a numerical-equivalence proof. This block is restricted to AERIS-versus-LEACH. Under aligned multi-environment settings, AERIS is directionally not lower than LEACH in all tested environment-scale cells. At 100 nodes, significance is confirmed in 3/4 environments (indoor_factory, outdoor_urban, outdoor_suburban), while indoor_office remains non-significant. In the 300/500-node extension, all 8 environment-scale comparisons are significant after Holm correction.

Table 5. NS-3 trend-level validation at 100 nodes (AERIS vs LEACH, n=30; metric=`pdr_expected`).

Environment	AERIS	LEACH	Delta	Holm-corrected p	Significance
indoor_office	0.9202	0.9185	+0.0017	1.00	no
indoor_factory	0.5991	0.5330	+0.0661	$< 10^{-20}$	yes
outdoor_urban	0.2057	0.1889	+0.0169	7.89×10^{-4}	yes
outdoor_suburban	0.7767	0.6929	+0.0838	3.17×10^{-25}	yes

Scope note: this NS-3 block supports trend consistency only. The manuscript does not claim cross-platform numerical equivalence. Across 5 node counts and 4 environments, 18/20 AERIS-versus-LEACH comparisons are significant.

6. Discussion

The evidence supports a scoped positioning rather than universal superiority. AERIS is strong for reliability across heterogeneous channels, but it is not the top protocol in every environment-scale pair. Indoor_office is a stable counterexample where PEGASIS is higher in the scalability matrix.

Practical interpretation requires separating statistical and engineering significance. At very large sample sizes, tiny absolute gaps can be statistically significant; therefore, conclusions in this paper rely on both corrected p values and absolute PDR deltas. This is particularly relevant for indoor_office at large scale, where ordering differences are small in magnitude.

Energy interpretation also requires caution: lower total energy can co-occur with shorter lifetime if a protocol terminates earlier. For this reason, reliability-energy statements are framed as conditioned trade-offs under fixed evaluation settings, not universal dominance claims.

The manuscript intentionally keeps three evidence blocks separate:

- 100-node multi-environment comparison,
- large-scale matrix (100–1000 nodes) with mixed environment-specific sample sizes,
- NS-3 trend validation for AERIS versus LEACH only.

Cross-block numerical pooling is avoided. This separation is deliberate: the 100-node block and the 100–1000-node block answer different questions and therefore should not be merged into a single headline number.

7. Limitations and Future Work

- Current evidence is simulator-based.
- NS-3 trend-level publication gate is completed; numerical equivalence is intentionally not claimed because of platform-depth differences.
- The large-scale matrix uses environment-specific sample sizes (n=550 or n=650), so conclusions are not presented as a pooled single-regime estimate.
- Reported gains are bounded to the tested baseline set and channel taxonomy.
- Hop count is reported as a route-length proxy, not as a wall-clock latency benchmark under full MAC scheduling.
- Additional topology families (corridor, hotspot) and hardware-in-the-loop validation remain pending.

8. Conclusion

AERIS provides a reliability-oriented routing option for heterogeneous WSN channels. In the publication-tier 100-node matrix, it achieves the highest PDR within the tested baseline set across all four environments. In the large-scale matrix, it remains first in three of four environments, with indoor_office as a stable counterexample where PEGASIS is higher. NS-3 confirms trend consistency for AERIS versus LEACH and is used strictly as trend-level evidence. Gateway effects are environment-dependent and CAS effects are mixed; therefore, the final claim is explicitly bounded to the tested protocol set, environments, metrics, and sample-size regimes.

Author Contributions: Conceptualization, X.Z. and K.L.; methodology, K.L. and J.L.; software, K.L.; validation, X.Z. and J.L.; writing—original draft preparation, K.L.; writing—review and editing, X.Z.; supervision, X.Z.

Funding: This research received no external funding.

Conflicts of Interest: The authors declare no conflict of interest.

Data Availability Statement

The datasets, scripts, manuscript sources, and provenance sidecar records (including commit identifiers and script hashes) supporting this study are available from the corresponding author and will be released in the public project repository upon final publication.

References

1. Akyildiz, I.F.; Su, W.; Sankarasubramaniam, Y.; Cayirci, E. A Survey on Sensor Networks. *IEEE Communications Magazine* **2002**, *40*, 102–114. doi:10.1109/MCOM.2002.1024422.
2. Rault, T.; Bouabdallah, A.; Challal, Y. Energy Efficiency in Wireless Sensor Networks: A Top-Down Survey. *Computer Networks* **2016**, *91*, 104–132. doi:10.1016/j.comnet.2015.07.014.
3. Kandris, D.; Nakas, C.; Vomvas, D.; Koulouras, G. Power Conservation through Energy Efficient Routing in Wireless Sensor Networks: A Survey. *Sensors* **2020**, *20*, 1940. doi:10.3390/s20071940.
4. Heinzelman, W.R.; Chandrakasan, A.; Balakrishnan, H. Energy-Efficient Communication Protocol for Wireless Microsensor Networks. Proceedings of the 33rd Annual Hawaii International Conference on System Sciences, 2000. doi:10.1109/HICSS.2000.926982.
5. Lindsey, S.; Raghavendra, C. PEGASIS: Power-Efficient Gathering in Sensor Information Systems. Proceedings, IEEE Aerospace Conference, 2002, Vol. 3, pp. 1125–1130. doi:10.1109/AERO.2002.1035242.
6. Younis, O.; Fahmy, S. HEED: A Hybrid, Energy-Efficient, Distributed Clustering Approach for Ad Hoc Sensor Networks. *IEEE Transactions on Mobile Computing* **2004**, *3*, 366–379. doi:10.1109/TMC.2004.41.
7. Manjeshwar, A.; Agrawal, D.P. TEEN: A Routing Protocol for Enhanced Efficiency in Wireless Sensor Networks. Proceedings 15th International Parallel and Distributed Processing Symposium, 2001, pp. 2009–2015. doi:10.1109/IPDPS.2001.925197.
8. Zuniga, M.; Krishnamachari, B. Analyzing the Transitional Region in Low Power Wireless Links. 2004 First Annual IEEE Communications Society Conference on Sensor and Ad Hoc Communications and Networks. IEEE, 2004, pp. 517–526. doi:10.1109/SAHCN.2004.1381954.
9. Zuniga Zamalloa, M.; Krishnamachari, B. An Analysis of Unreliability and Asymmetry in Low-Power Wireless Links. *ACM Transactions on Sensor Networks* **2007**, *3*, 7. doi:10.1145/1240226.1240227.
10. Baccour, N.; Koubaa, A.; Mottola, L.; Zuniga, M.A.; Youssef, H.; Boano, C.A.; Alves, M. Radio Link Quality Estimation in Wireless Sensor Networks. *ACM Transactions on Sensor Networks* **2012**, *8*, 1–33. doi:10.1145/2240116.2240123.
11. Heinzelman, W.; Chandrakasan, A.; Balakrishnan, H. An Application-Specific Protocol Architecture for Wireless Microsensor Networks. *IEEE Transactions on Wireless Communications* **2002**, *1*, 660–670. doi:10.1109/TWC.2002.804190.
12. Ren, J.; Zhang, Y.; Li, D.; Wang, J. MeFi: Mean Field Reinforcement Learning for Cooperative Routing in Wireless Sensor Networks. *IEEE Internet of Things Journal* **2024**, *11*, 995–1011. doi:10.1109/JIOT.2023.3294826.

13. Okine, A.A.; Adam, N.; Naeem, F.; Kaddoum, G. Multi-Agent Deep Reinforcement Learning for Packet Routing in Tactical Mobile Sensor Networks. *IEEE Transactions on Network and Service Management* **2024**, *21*, 2155–2169. doi:10.1109/TNSM.2023.3321456.
14. Chen, L.; Yang, J.; Zhou, W. A Comprehensive Survey on Machine Learning for Wireless Sensor Networks: Applications, Challenges, and Future Directions. *IEEE Communications Surveys & Tutorials* **2023**, *25*, 1567–1612. doi:10.1109/COMST.2023.3267890.
15. Liu, X.; Zhao, J.; Wang, X. Environment-Adaptive Routing in Wireless Sensor Networks Based on Machine Learning. *Sensors* **2018**, *18*, 3716. doi:10.3390/s18113716.
16. Zhao, S.; Wang, Y.; Sun, L. Context-Aware Routing Protocols for Vehicular Sensor Networks: A Survey. *Computer Networks* **2019**, *155*, 153–168. doi:10.1016/j.comnet.2019.03.021.